

HTML5 Semantic Tags, Forms, Validation, Audio and Video

Introduction to HTML5

HTML stands for:

HyperText Markup Language

HTML is used to create the structure of web pages.

HTML5 is the modern version of HTML.

HTML5 provides many useful features such as:

- Semantic elements
- Form validation
- Audio
- Video
- Graphics
- Better support for modern web applications

What is Semantic HTML?

Semantic HTML means using HTML elements according to their **meaning**.

For example:

```
<header>
```

is used for the header section of a webpage.

```
<footer>
```

is used for the footer section.

```
<nav>
```

is used for navigation links.

The HTML element itself tells us what the content represents.

Why Use Semantic Tags?

Before HTML5, developers often used many `<div>` elements.

Example:

```
<div>
  Website Header
</div>

<div>
  Navigation
</div>

<div>
  Main Content
</div>

<div>
  Footer
</div>
```

The browser can display this, but it is difficult to understand the purpose of each `<div>`.

With semantic HTML:

```
<header>
  Website Header
</header>

<nav>
  Navigation
</nav>

<main>
  Main Content
</main>

<footer>
  Footer
</footer>
```

Now the structure is much easier to understand.

Advantages of Semantic HTML

Semantic HTML provides several benefits.

1. Readability

The code becomes easier for developers to understand.

2. Accessibility

Screen readers can better understand the structure of the webpage.

3. SEO

Search engines can better understand the structure and meaning of webpage content.

4. Maintainability

It becomes easier to modify and maintain the webpage.

`<header>` Tag

The `<header>` element represents introductory content for a webpage or a section.

It commonly contains:

- Website name
- Logo
- Heading
- Introduction
- Navigation

Example:

```
<header>
```

```
  <h1>My Website</h1>
```

```
  <p>Welcome to my website</p>
```

```
</header>
```

<nav> Tag

The `<nav>` element contains navigation links.

Example:

```
<nav>

  <a href="index.html">Home</a>

  <a href="about.html">About</a>

  <a href="contact.html">Contact</a>

</nav>
```

The `<nav>` element tells the browser and assistive technologies that these links are used for navigation.

<main> Tag

The `<main>` element contains the main content of the webpage.

Example:

```
<main>

  <h2>Welcome</h2>

  <p>This is the main content of the website.</p>

</main>
```

A webpage normally has one main `<main>` element.

<section> Tag

The `<section>` element represents a logical section of content.

Example:

```
<section>

  <h2>About Us</h2>

  <p>
    We provide web development courses.
  </p>

</section>
```

A section normally has a heading.

An article should make sense as an independent piece of content.

`<footer>` Tag

The `<footer>` element represents the footer of a webpage or section.

It commonly contains:

- Copyright information
- Contact information
- Additional links
- Author information

Example:

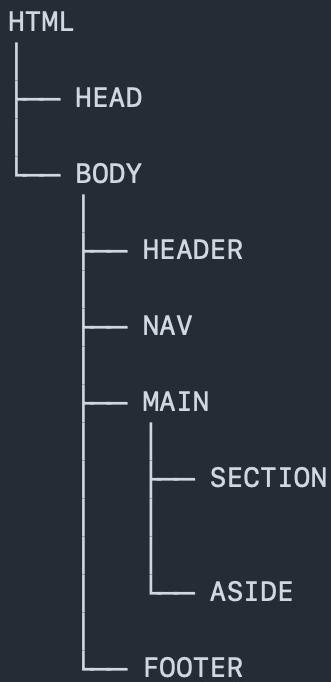
```
<footer>

  <p>Copyright 2026 My Website</p>

</footer>
```

Understanding the Structure

The structure can be visualized as:



HTML Form Validation

Validation means checking whether the information entered by the user is valid.

For example:

A required name should not be empty.

An email should have a valid email format.

A phone number should contain the expected number of digits.

A password should follow specific rules.

HTML5 provides several validation attributes.

Important validation attributes include:

```
required  
type  
pattern  
minlength  
maxlength  
min  
max
```

The **required** Attribute

The `required` attribute makes a form field compulsory.

Example:

```
<input type="text" required>
```

The user must enter a value before submitting the form.

Example of `required`

```
<form>

  <label for="name">Name</label>

  <input type="text"
        id="name"
        required>

  <br><br>

  <button type="submit">Submit</button>

</form>
```

If the user leaves the name empty and clicks Submit, the browser displays a validation message.

The `pattern` Attribute

The `pattern` attribute is used to specify a custom validation pattern.

It uses a regular expression.

Example:

32. The ``minlength`` Attribute

The ``minlength`` attribute specifies the minimum number of characters allowed.

Example:

```
```html
<input type="text"
```

```
minlength="5">
```

The user must enter at least 5 characters.

# `minlength` and `maxlength`

Example:

```
<input type="text"
 minlength="5"
 maxlength="20">
```

This means:

Minimum = 5 characters  
Maximum = 20 characters

## The `min` Attribute

The `min` attribute specifies the minimum numerical value.

Example:

```
<input type="number"
 min="18">
```

The value should not be less than 18.

## The `max` Attribute

The `max` attribute specifies the maximum numerical value.

Example:

```
<input type="number"
 max="60">
```

The value should not be greater than 60.



# Using `min` and `max` Together

Example:

```
<input type="number"
 min="18"
 max="60">
```

The accepted range is:

18 to 60

## `minlength` / `maxlength` vs `min` / `max`

These attributes are different.

Attribute	Used For
<code>minlength</code>	Minimum number of characters
<code>maxlength</code>	Maximum number of characters
<code>min</code>	Minimum numerical value
<code>max</code>	Maximum numerical value

Example:

```
<input type="text"
 minlength="5"
 maxlength="20">
```

Character length is controlled.

Example:

```
<input type="number"
 min="18"
 max="60">
```

Numerical value is controlled.

# HTML5 Multimedia

HTML5 provides built-in support for multimedia.

The two important multimedia elements are:

```
<audio>
```

and

```
<video>
```

These allow audio and video to be embedded directly into webpages.

## Audio in HTML

The `<audio>` element is used to embed audio content.

Examples:

- Music
- Podcasts
- Voice recordings
- Audio lessons
- Sound effects

## The `<audio>` Tag

Basic syntax:

```
<audio>

 <source src="song.mp3"
 type="audio/mpeg">

</audio>
```

However, without controls, the user may not have a way to play the audio.

Therefore, we commonly use:

```
<audio controls>

 <source src="song.mp3"
 type="audio/mpeg">

</audio>
```

## The `controls` Attribute

The `controls` attribute displays audio controls.

Example:

```
<audio controls>

 <source src="song.mp3"
 type="audio/mpeg">

</audio>
```

The browser can display controls such as:

- Play
- Pause
- Volume
- Progress bar

## The `autoplay` Attribute

The `autoplay` attribute asks the browser to start playing the audio automatically.

Example:

```
<audio controls autoplay>

 <source src="song.mp3"
 type="audio/mpeg">
```

```
</audio>
```

Important:

Modern browsers often restrict autoplay when audio has sound because automatic playback can be annoying to users.

Therefore, autoplay should be used carefully.

## The `loop` Attribute

The `loop` attribute causes the audio to play again after it finishes.

Example:

```
<audio controls loop>

 <source src="song.mp3"
 type="audio/mpeg">

</audio>
```

The audio repeats continuously.

## Combining Audio Attributes

We can use:

```
controls
autoplay
loop
```

together.

Example:

```
<audio controls autoplay loop>

 <source src="song.mp3"
 type="audio/mpeg">

</audio>
```

This means:

`controls` → Show audio controls

`autoplay` → Try to start automatically

`loop` → Repeat after finishing

## Video in HTML

The `<video>` element is used to embed video content.

Examples:

- Educational videos
- Tutorials
- Product demonstrations
- Training videos
- Presentations

## The `<video>` Tag

Basic example:

```
<video controls>

 <source src="video.mp4"
 type="video/mp4">

</video>
```

## The `controls` Attribute for Video

The `controls` attribute displays video controls.

Example:

```
<video controls>

 <source src="video.mp4"
 type="video/mp4">

</video>
```

The browser may provide:

- Play
- Pause
- Volume
- Progress bar
- Fullscreen

## The **poster** Attribute

The **poster** attribute specifies an image that is displayed before the video starts playing.

Example:

```
<video controls
 poster="thumbnail.jpg">

 <source src="video.mp4"
 type="video/mp4">

</video>
```

The image:

thumbnail.jpg

acts as a preview image for the video.

## Why Use **poster**?

A poster image can provide:

- A preview of the video
- A professional appearance

- Information about the video
- Better user experience

Example:

